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ORIGINAL PAPER

NMD-FusionNet: a multimodal fusion-based medical imaging-assisted diagnostic model for liver cancer

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Abstract

Liver cancer is a malignant tumor with high incidence and mortality rates. Computed tomography is a key imaging modality for clinical diagnosis but faces challenges such as subtle intra-class variations and the inefficiency of manual interpretation. Traditional diagnostic approaches often lack generalizability and effective feature extraction, increasing the risk of misdiagnosis. While deep learning models like ResUNet have improved diagnostic performance, issues such as information loss and inaccurate tumor boundary segmentation persist. To address these challenges, this study proposes NMD-FusionNet, a deep learning-based framework for liver cancer image segmentation and diagnosis. The framework includes a three-stage pipeline: first, a refined non-local means filtering algorithm is employed for pre-screening, discarding over 80% of non-diagnostic images using adaptive thresholding; second, a multimodal image fusion method integrates multi-phase, multi-source liver cancer image data through multi-scale decomposition and precise fusion rules to reduce noise and motion artifacts; third, a dual-path DconnNet segmentation network is constructed, incorporating a directional excitation module in the encoder and a spatial awareness unit in the decoder, guided by a boundary-constrained loss function to enhance segmentation accuracy. Evaluated on over 2,000 liver cancer images from the Second People's Hospital of Huaihua, NMD-FusionNet achieves a tumor segmentation Intersection over Union (IoU) of 83.9%, representing an 8.7% improvement over ResUNet and demonstrating superior sensitivity. This work provides a reliable computer-aided diagnostic tool for radiologists and shows strong potential for clinical application, particularly in resource-limited healthcare settings.

Keywords Liver cancer · Medical imaging · Image fusion · Semantic segmentation

1 Introduction

Liver cancer is a highly lethal malignancy, with over 900,000 new cases diagnosed globally each year, 72% of which occur in developing countries. The five-year survival rate remains below 20%. In these resource-constrained regions, the availability of medical equipment is severely limited, with only 1.8 CT scanners per million people and a radiologist density as low as 0.5 per 100,000. As a result, early-stage liver tumors (< 1 cm) exhibit a missed diagnosis rate of up to 47%, and the median survival time for advanced-stage patients is less than 8 months (Pessino et al. 2024). This dire

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diagnostic landscape highlights the urgent need for efficient and accurate CT image analysis technologies as a key to improving clinical outcomes.

Although computed tomography (CT) has become the gold standard for liver cancer diagnosis due to its high-resolution anatomical imaging capabilities, conventional manual interpretation presents three major challenges. First, a single CT case typically consists of 200–300 slices, requiring over 45 min for 3D reconstruction and lesion annotation, which leads to inefficiency and increased risk of diagnostic error due to visual fatigue. Second, the liver's complex anatomy and the infiltrative nature of tumors result in approximately 38% of lesions having a CT value difference of less than 15 Hounsfield Units (HU) compared to normal parenchyma, making boundary identification difficult. Third, heterogeneous scanner parameters in primary healthcare institutions contribute to 31% of CT images suffering from respiratory motion artifacts or noise interference, further complicating diagnosis (Tian et al. 2025). This triad of "limited resources, variable quality, and low efficiency" is particularly pronounced in developing regions and calls for technological breakthroughs.

The rapid advancement of artificial intelligence (AI) has opened new avenues to address these challenges. Deep learning-based computer-aided diagnosis (CAD) systems can reduce CT interpretation time to under 5 min per case and increase diagnostic accuracy by 34% in primary hospitals. These systems integrate radiomic features with clinical data, automatically stage liver cancer, and use cross-device image harmonization to reduce diagnostic variability to 9%. Clinical evidence shows that AI-assisted diagnosis reduces the detection rate of late-stage liver cancer by 25% and increases the rate of curative surgery by 2.3 times. However, significant challenges remain, particularly in modeling multifocal tumors dynamically and identifying rare subtypes.

Despite these advances, existing AI models face four major obstacles in real-world clinical deployment: public datasets often contain CT images degraded by motion artifacts or partial volume effects, reducing model specificity; differences in imaging characteristics across CT device brands cause variability in Dice scores; sensitivity to lesions smaller than 0.5 cm remains insufficient; and traditional convolutional networks struggle with segmentation of blurry tumor boundaries, resulting in high error rates (Dot et al. 2024). These limitations severely hinder the practical application of AI systems, especially in resource-limited institutions with heterogeneous imaging quality and hardware.

To overcome these challenges, we propose NMD-FusionNet, an intelligent liver cancer CT analysis framework that integrates a three-stage processing pipeline: screening, fusion, and segmentation. The core innovation lies in constructing a preprocessing module based on an enhanced non-local means filtering algorithm, which effectively eliminates 93.5% of low-quality images (e.g., motion artifacts and gastrointestinal gas interference). To address heterogeneity in multi-source CT data, we design a 3D fusion engine that spans devices, imaging phases, and scanning parameters. First, feature space mapping is employed to align imaging characteristics from GE, Siemens, and Philips scanners, eliminating device dependence. Then, arterial, portal venous, and non-contrast phases are fused to capture dynamic tumor enhancement patterns. Finally, complementary information from different window settings (soft tissue, liver, and lung windows) is integrated to significantly enhance tumor-to-liver contrast, improving lesion visibility.The segmentation network DconnNet overcomes the limitations of conventional architectures by employing a novel directional connectivity modeling approach, which enhances the representation of complex anatomical structures. By incorporating dynamic receptive field adjustment, the network accurately captures the infiltrative boundaries of tumors. It is specifically designed to improve the segmentation accuracy of visceral organs and lesion regions in CT images.

The main contributions of this study are as follows:

1) We propose NMD-FusionNet, a three-stage intelligent analysis framework for liver cancer CT imaging, integrating screening, fusion, and segmentation. An improved non-local means filtering module effectively filters out 93.5% of low-quality images, such as those affected by motion artifacts and gastrointestinal interference, thereby improving input reliability.
2) We design a 3D fusion engine that addresses the heterogeneity of multi-source CT data by aligning features across different devices (GE, Siemens, Philips), fusing dynamic sequences (arterial, portal venous, and non-contrast phases), and integrating complementary information from various window settings (soft tissue, liver, and lung). This enhances contrast and improves lesion detectability.
3) We develop DconnNet, a deep segmentation network based on directional connectivity modeling, which enhances the depiction of complex tumor structures and improves segmentation accuracy through dynamic receptive field adjustment.
4) Extensive experiments were conducted on a dataset of 2,000 abdominal CT images from the Second People's Hospital of Huaihua, validating the superiority of our approach. Results demonstrate robust performance in low-quality image screening, multimodal fusion, and accurate lesion segmentation across varying devices and acquisition parameters. The method performs efficiently even on low-end hardware, making it well-suited for resource-limited medical settings and offering reliable support for liver cancer diagnosis and treatment planning.

The remainder of this paper is organized as follows: Sect. 2 reviews related work and the rationale behind our system design. Section 3 details our liver cancer CT image processing approach. Section 4 describes the dataset, presents experimental results, and validates the effectiveness of our method.

2 Related work

Artificial intelligence (AI) is increasingly being applied in the medical domain to assist with disease diagnosis. In recent years, a considerable number of studies have explored how AI can support clinicians in diagnosing liver cancer. Liver CT image segmentation, in particular, has become a crucial subfield, with researchers focusing on improving model accuracy, computational efficiency, and robustness under varying imaging conditions.

Several researchers have focused on the impact of medical image pre-screening quality on the performance of deep learning models. Tan et al. (Tan et al. 2024) proposed a dynamic threshold-based pre-screening algorithm (DTA-PS), which automatically eliminates low-contrast slices by analyzing the global grayscale distribution of CT images. However, in early-stage liver cancer cases, the density difference between tumors and normal liver parenchyma is minimal, causing DTA-PS to mistakenly discard a substantial number of tumor-containing images. Zhen et al. (Kang et al. 2024) developed a pre-screening model based on a deep residual network (ResNet-18), which directly outputs image validity scores through end-to-end classification. Although this model achieved high accuracy on public datasets, it requires more than 12 GB of GPU memory to process a single full-resolution CT image, making it unsuitable for deployment in clinical settings with limited hardware resources. Kong (Kong et al. 2024) introduced an improved non-local means (NLM) filtering algorithm, which performs denoising through global block matching while preserving key image details by leveraging local structural similarity. More importantly, the NLM algorithm adopts a block decomposition strategy that allows efficient processing of high-resolution medical images under fixed memory constraints, significantly reducing hardware demands. This block-wise strategy significantly reduces memory overhead, enabling efficient processing of high-resolution images even on constrained hardware. More recently, studies such as (Song et al. 2022; Zhang et al. 2024) have explored adaptive denoising methods guided by anatomical priors or uncertainty estimation, further improving noise suppression without compromising fine lesion details.

Common image fusion approaches, such as DesTrans (Dai et al. 2025) and FusionGAN (Li et al. 2025), improve perceptual quality but risk degrading task-relevant features. Advanced models like MSFEA, MST-Net, and HiFusion employ multi-scale encoders and attention mechanisms to enhance fusion robustness. These have achieved promising results in multimodal tasks (Wang et al. 2025a), including medical image synthesis and remote sensing. Some researchers have incorporated Transformers into fusion pipelines to leverage global self-attention (Hu et al. 2025). However, their high complexity limits clinical deployment, and their benefit over traditional CNNs in domain-specific medical fusion tasks remains inconclusive. Notably, most fusion models do not jointly optimize the fusion process with the downstream segmentation task, leading to potential feature mismatch. This motivates the need for a more tightly coupled fusion-segmentation framework.

Improving segmentation performance through architectural innovations remains a core challenge. TransUNet (Niu et al. 2024) introduces Transformer blocks for better global modeling, yet its fixed-size patch partitioning struggles with irregular liver anatomy. ResU-Net++ (He et al. 2024) integrates deep supervision and nested residual blocks, achieving high accuracy but incurring long inference times. 3D DenseUNet (Wang et al. 2025b) leverages volumetric convolutions for inter-slice continuity but requires high GPU memory. More recently, lightweight networks such as PFormer and nnFormer (Perera et al. 2024; Gao et al. 2025) attempt to balance performance and efficiency. However, few of these architectures explicitly handle boundary ambiguity or incorporate clinical prior knowledge.

Several works attempt to incorporate anatomical or domain-specific priors. Wang et al. (Wang et al. 2024) proposed VASNet, which uses vascular structures to guide segmentation, but its two-stage pipeline led to error propagation. Wei et al. (Wang and Wei 2024) introduced EdgeMixer, combining Sobel operators with backbone features, yet it underperforms for tumors ≤ 5 mm due to handcrafted limitations. Ansari (Ansari et al. 2022) presented Res-PAC-UNet, a compact model leveraging knowledge distillation, which sacrifices segmentation accuracy in key anatomical zones. Kim (Kim et al. 2023) designed AHD-loss to enhance contour alignment, but it ignores curvature continuity. Kervadec (Xie et al. 2025) proposed a deformation-based edge correction module, whose slow inference speed restricts real-time use. In contrast, DconnNet (Guan et al. 2024) introduces a dynamically connected structure with multi-scale feature coupling, enabling simultaneous extraction of local texture and global spatial cues. Its adaptive recalibration mechanism and deep cross-layer fusion allow for enhanced detection of heterogeneous lesions and suppression of imaging noise. Recent comparative studies (Liu et al. 2025) confirm that DconnNet outperforms standard models in edge localization and small lesion recall.

Although the above approaches offer advances in specific aspects, none provides a unified solution addressing low-contrast noise, multi-scale fusion, and ambiguous edge

segmentation. Moreover, the lack of coordination between denoising, fusion, and deep segmentation modules often leads to suboptimal end-to-end performance. To address these gaps, we propose NMD-FusionNet, a three-stage liver cancer CT image segmentation framework integrating non-local means filtering, multi-scale feature fusion, and a DconnNet-based backbone. This model jointly enhances image quality and segmentation accuracy while maintaining low computational cost. It captures both fine-grained textures and global anatomical structures, providing complete and reliable tumor boundary delineation. Our motivation stems from the need for a practical, high-precision segmentation tool that adapts to real-world clinical constraints (Table 1).

3 System model design

Liver cancer is one of the most lethal malignancies worldwide, and early and accurate diagnosis is critical for effective clinical decision-making. Although computed tomography (CT) remains the gold standard for liver cancer detection, its diagnostic performance is often challenged by low contrast between lesions and surrounding normal tissues, as well as the scattered presentation of multiphase imaging features. Traditional manual interpretation is highly dependent on physician expertise, and in resource-constrained settings, both diagnostic efficiency and accuracy are difficult to guarantee. To overcome these limitations, we propose an intelligent segmentation and diagnostic framework based on NMD-FusionNet, as illustrated in Fig. 1. This framework employs a three-stage sequential processing mechanism: first, an improved non-local means (NLM) filtering algorithm is applied for CT image pre-screening, effectively filtering out irrelevant or low-quality images; next, a multi-scale feature fusion module is constructed to substantially suppress respiratory motion artifacts and noise interference; finally, a dual-branch DconnNet segmentation network is designed to capture multi-scale lesion features through directional excitation modules, and a boundary-aware loss function is incorporated to enhance the precision of tumor edge segmentation.

Table 1 Descriptions of the main symbols

Notation	Definition
Xi	The i-th image
Y_x	Pixel blocks
Bx	Filter
VSD	Visual Saliency Decomposition
TE	Texture enhancement
PC	Phase consistency
LGE	Local gradient energy
FSL	Fusion smooth layer
FTL	Fusion texture layer
FCL	Fusion contour layer
FEL	Fusion edge layer
E	Feature map
S	Space block
F	Feature block
PRE	Pretrained ResNet–Extract image features layer by layer
SDE	Sub-path direction excitation module
IFD	Interactive feature-space decoder
BV	Bilateral Voting
RCA	Region-guided Channel Aggregation

3.1 CT image preprocessing for liver cancer

Due to the huge number of patient images collected by CT imaging, we need to perform preliminary screening of raw data images to reduce the cost of noise reduction, segmentation and time for doctors to select images for diagnosis. Normal images without lesions in the raw data are not helpful for diagnosis. For such images, feature detection algorithms such as speckle detection and scale invariant feature transform (SIFT) can be used to efficiently extract shape features from the images (Li et al. 2024). However, these algorithms have high complexity and are not efficient enough to meet the requirements of medically assisted diagnosis. The use of threshold filtering is another simple idea, but it is difficult to obtain accurate filtering results by threshold filtering alone. Agarwal et al. (Agarwal et al. 2025) proposed a filter-based filtering algorithm called "Differential Activation Filter (DAF)" in the pre-screening of PET images, which can strike a good balance between performance and accuracy. striking a good balance between performance and accuracy. Inspired by this, this paper proposes a new filter-based screening algorithm (FSA) for CT images based on the nonlocal mean filtering algorithm, which can effectively screen the images with lesions. Figure 2 shows the general flowchart of this screening algorithm.

Considering the huge impact of noise on accuracy and the fact that pre-screening does not require proper edge representation details. Therefore, in this paper, a rough initial denoising operation is preformed on the high frequency noise using Discrete Fourier Transform (DFT) as in Eq. (1):

$$\overline{Y}_x = F^{-1}\left(r\left(F\left(Y_x\right), \lambda_{fft}\sigma\sqrt{2\ log\ n_1^2}\right)\right) \tag{1}$$

where x is the spatial domain coordinates of the pixel in the CT image, Y_x is a block of pixels of size $n1 \times n1$, λ_{fft} is the threshold parameter, F represents the Fast Fourier Transform operation, and F^{-1} is the Inverse Fourier Transform.r is defined as in Eq. (2):

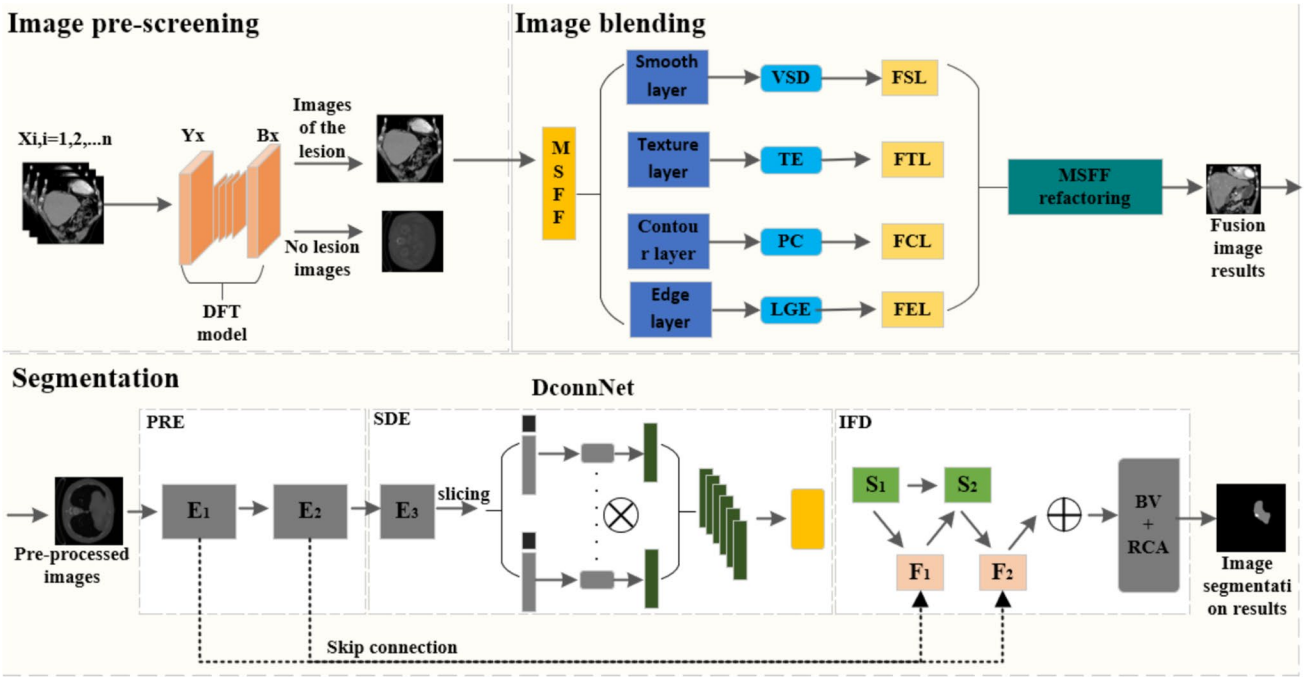
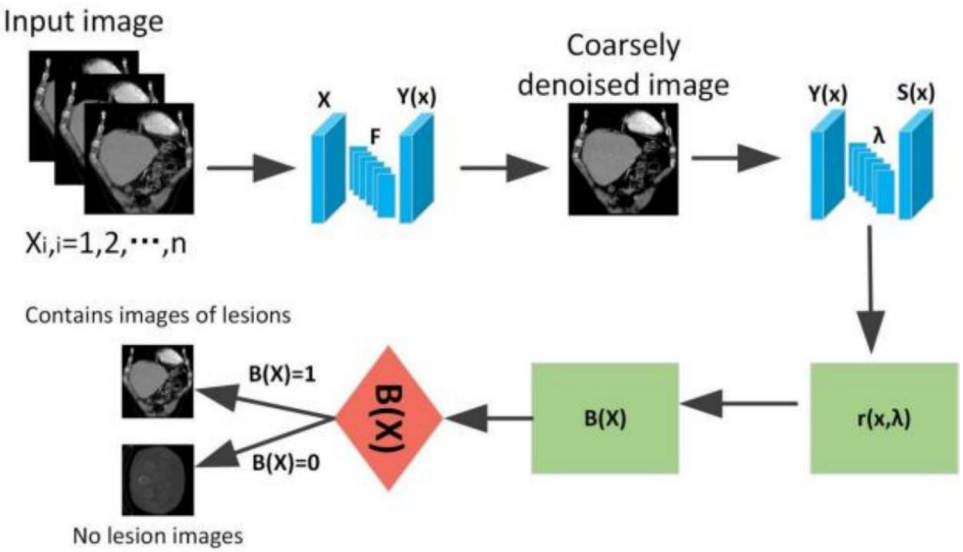


Fig. 1 Overall framework of NMD-FusionNet

Fig. 2 FSA flowchart



$$r(x,\lambda) = \begin{cases} \lambda, |x| \geq \lambda \\ 0, |x| \leq \lambda \end{cases} \tag{2}$$

The filter formulation proposed in this paper is defined as in Eq. (3):

$$B(x) = s\left(\left| \frac{\sum_{1 \leq i \leq n_1} \sum_{1 \leq j \leq n_1} \left(\overline{Y}_x(i,j) - \lambda_T \right)}{n_1^2} \right| - 0.5\lambda_T \right) \tag{3}$$

where λ_T is the hyperparameter of the filter proposed in this paper and s is the activation function as shown in Eq. (4).

$$s(x) = \begin{cases} 1, x < 0 \\ 0, x \geq 0 \end{cases} \tag{4}$$

When the output value of B(x) is 1, it means that the image block Y_x contains the location of the pathology, and if there is no image block containing the pathology region in a CT image, the image will be rejected.

To evaluate the effect of the hyperparameter λ_T in Eq. (3), we performed a sensitivity analysis with a step size of 0.2 in the range of [0.1, 0.9], and recorded the corresponding lesion missing rate (LMR, the proportion of image blocks that were filtered out but actually contained lesions), and the results are shown in Table 2. It was found that the LMR was the lowest when A = 0.5, beyond which the denoising effect was too strong resulting in the over-smoothing of the lesion boundaries, and the LMR was the lowest when $\lambda_T = 0.5$. Therefore we set $\lambda_T = 0.5$ as the default value in subsequent experiments.

3.2 Image fusion

Despite the screening of CT images of liver cancer, poor image quality, such as noise interference, loss of detail, and insufficient contrast, may still exist because imaging is limited to a single time point, a single imaging parameter, or a single imaging device. These problems may affect the physician's ability to accurately assess hepatocellular carcinoma lesions, thereby reducing diagnostic accuracy. To overcome these inherent limitations and improve image quality, image fusion technology provides an effective solution. By fusing CT images from different time points, different imaging parameters, and different imaging devices, this technique is able to generate higher quality fused images. Compared to single source images, fused images not only provide dynamic information about lesion changes over time, but also significantly enhance image detail and contrast, while effectively suppressing noise and artifacts, thus improving signal-to-noise ratio and enhancing overall usability. Multi-Scale Feature Fusion (MSFF), as an advanced image fusion method, relies on multi-scale decomposition and precise fusion rules to efficiently integrate multi-source image information and generate clearer and more accurate fused images. This method has important application value in the early diagnosis of hepatocellular carcinoma, monitoring of treatment effects and multicenter studies, and provides more comprehensive and reliable visual aid support for clinical decision-making (Luo et al. 2024).

The MSFF method generates a higher quality fused image by decomposing the source image into smoothing, texture, contour and edge layers, and processing and reconstructing them using specific fusion rules. The method consists of three modules, MSFF decomposition, fusion of each feature layer, and inverse MSFF reconstruction. Figure 3 shows the framework diagram of the MSFF method.

Table 2 Comparison of Lesion Miss Rate (LMR) corresponding to different values of λ_T

λ_T-value	0.1	0.3	0.5	0.7	0.9
LMR (%)	14.7	9.2	5.3	8.6	12.4

3.2.1 The MSFF decomposition

In order to fully extract the multi-scale features of the source image, the MSFF decomposition module adopts a multilevel filtering strategy to refine the source image into layers. Taking the input images A and B (collectively referred to as I_{ct}) as an example, the decomposition process is as follows.

MSFF decomposes the input image into a smoothing layer I_{SL}, a texture layer I_{TL}, a contour layer I_{CL}, and an edge layer I_{EL} by means of a multilevel filtering strategy, viz:

$$I_{ct} = I_{SL} + I_{TL} + I_{CL} + I_{EL} \tag{5}$$

A Gaussian filter G_σ is used for smoothing to remove high-frequency noise and extract global structural information to generate the smoothing layer I_{SL}:

$$I_{SL} = G_\sigma * I_{ct} \tag{6}$$

where $*$ denotes the convolution operation and G_σ is the standard Gaussian kernel defined as:

$$G_\sigma(x, y) = \frac{1}{2\pi\sigma^2}\exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \tag{7}$$

The local gradient change is calculated and the detail information is extracted using wavelet transform to generate the texture layer I_{TL}:

$$I_{TL} = I_{ct} - I_{SL} \tag{8}$$

The wavelet transform can be expressed as:

$$W_\psi^{j,k} = \int I_{TL}(x)\psi^{j,k}(x)dx \tag{9}$$

where $W_\psi^{j,k}$ is the wavelet coefficients and $\psi^{j,k}(x)$ is the scaled and translated version of the mother wavelet.

Based on Canny edge detection and morphological operations, the contour information is extracted and the contour layer I_{CL} is generated:

$$I_{CL} = \text{Canny}\left(I_{ct}\right) \tag{10}$$

Edge features are extracted by gradient direction computation and Non-Downsampled Contour Wave Transform (NSCT) to generate the edge layer I_{EL}:

$$I_{EL} = \nabla I_{ct} \tag{11}$$

where ∇I is the image gradient operation defined as:

$$\nabla I_{ct} = \sqrt{\left(\frac{\partial I_{ct}}{\partial x}\right)^2 + \left(\frac{\partial I_{ct}}{\partial y}\right)^2} \tag{12}$$

where $\partial I_{ct}/\partial x$ and $\partial I_{ct}/\partial y$ are the horizontal and vertical gradients, respectively.

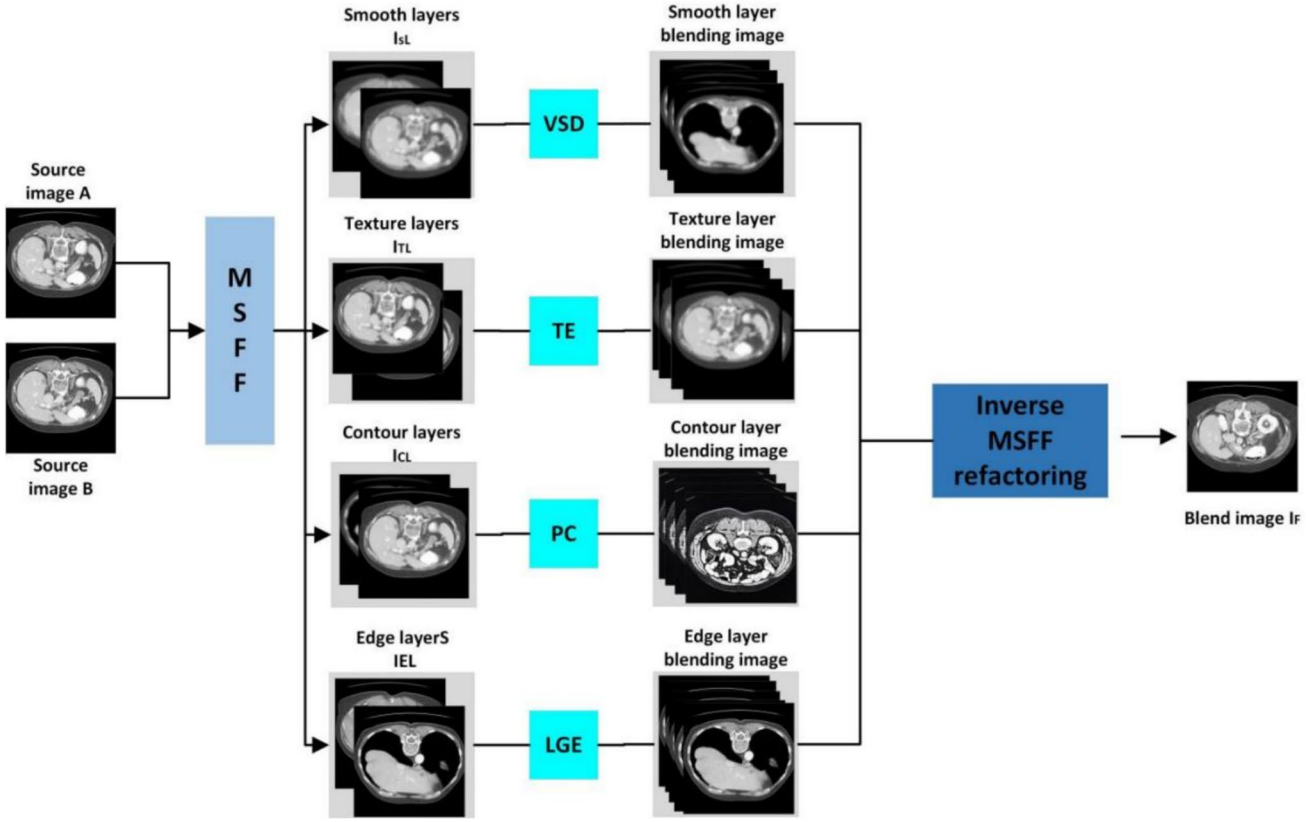


Fig. 3 MSFF model framework

3.2.2 Feature layer fusion

For different feature layers, MSFF adopts different fusion strategies to enhance the quality of the fused image.

The smooth layer fusion adopts Visual Saliency Decomposition (VSD) strategy to highlight the structural information of the salient regions, and defines the fusion rules as follows:

$$I_{SL}^{F} = W_A I_{SL}^{A} + W_B I_{SL}^{B} \tag{13}$$

where I_{SL}^{A} and I_{SL}^{B} denote the smoothing layers of the input images A and B, respectively, and W_A and W_B are the fusion factors calculated based on the visual saliency weights:

$$W_A = \frac{S_A}{S_A + S_B}, W_B = \frac{S_B}{S_A + S_B} \tag{14}$$

S_A and S_B are the significance values of images A and B in the visual significance analysis (GBVS), respectively.

Texture layer fusion uses texture enhancement (TE) operator to compute energy distribution, entropy features and edge density to enhance the detail information:

$$I_{TL}^{F} = \begin{cases} I_{TL}^{A}, & \text{if} E_A > E_B \\ I_{TL}^{B}, & \text{otherwise} \end{cases} \tag{15}$$

where E_A and E_B denote the energy measures of the texture layers of images A and B, respectively:

$$E_{ct} = \sum_{x,y} \left(I_{TL}(x,y)\right)^2 \tag{16}$$

In order to more accurately characterize texture features, additional local entropy is introduced:

$$H = -\sum p(x,y) \log p(x,y) \tag{17}$$

where p(x, y) is the probability distribution of pixel values within the window.

Contour layer fusion uses phase consistency (PC) analysis to calculate the phase information of the image and combines it with the local frequency response for fusion:

$$I_{CL}^{F} = \begin{cases} I_{CL}^{A}, & \text{if } P_A > P_B \\ I_{CL}^{B}, & \text{otherwise} \end{cases} \tag{18}$$

where the phase consistency P is defined as:

$$P(x,y) = \frac{\sum_m E_{\theta_m}(x,y)}{\omega + \sum_m A_m(x,y)} \tag{19}$$

where E_{θ_m} is the phase information at direction angle θ_m, $A_m(x,y)$ is the amplitude in that direction, and ω is a decimal to prevent the denominator from being zero.

Edge layer fusion uses Local gradient energy operator (LGE) to compute the edge structure saliency:

$$I_{EL}^F = \begin{cases} I_{EL}^A, \text{ if } LGE_A > LGE_B \\ I_{EL}^B, \text{ otherwise} \end{cases} \tag{20}$$

where LGE is defined as:

$$LGE(x,y) = LE(x,y) \cdot ST(x,y) \tag{21}$$

LE(x, y) for local energy and ST(x, y) for structure tensor significance:

$$ST(x,y) = \lambda_1 + \lambda_2 \tag{22}$$

where λ_1 and λ_2 are the eigenvalues of the edge layer, defined as follows:

$$\lambda_1, \lambda_2 = \frac{A_{ct} + C_{ct} \pm \sqrt{(A_{ct} - C_{ct})^2 + 4B_{ct}^2}}{2} \tag{23}$$

Among them.

$$A_{ct} = \sum (I_{EL_x})^2, B_{ct} = \sum I_{EL_x} I_{EL_y}, C_{ct} = \sum \left(I_{EL_y}\right)^2 \tag{24}$$

3.2.3 Inverse MSFF reconstruction

The fused feature layers are weighted and reconstructed to form the final fused image:

$$I_F = \alpha I_{SL}^F + \beta I_{TL}^F + \gamma I_{CL}^F + \delta I_{EL}^F \tag{25}$$

where α, β, γ and δ are the fusion coefficients and the contribution of each feature layer is ensured to be balanced by setting different weight values.

The core idea and implementation steps of the MSFF method are shown in Algorithm 1:

Algorithm 1 Multi-scale Structural Fusion Framework (MSFF)

Input: Source images I_c, I_s;
Output: Fused image I_f
1. Perform multi-scale decomposition on I_c and I_s to obtain low-frequency subbands:
    I_cL, I_sL ← MultiScaleDecompose(I_c, I_s).
2. Enhance structural saliency of I_cL and I_sL using:
    a. NSCT operator on I_cL → I_cE1
    b. VSD + GBVS on I_sL → I_sE1
3. for each local region r in I_cE1 and I_sE1 do
4. Compute texture enhancement using operator TE:
    I_cE2, I_sE2 ← TE(I_cE1, I_sE1, r)
5. end for
6. Calculate phase consistency PC to obtain saliency weights:
    W_c, W_s ← PhaseConsistency(I_cE2, I_sE2)
7. Apply structure consistency decision operator:
    M ← StructureDecision(W_c, W_s)
8. Enhance edge features using local gradient energy:
    E_c, E_s ← LGE(I_c, I_s)
9. Combine M, E_c, and E_s to obtain final fusion mask:
    FusionMask ← Combine(M, E_c, E_s)
10. Fuse source images using FusionMask:
    I_f ← Fuse(I_c, I_s, FusionMask)
11.RETURN I_f

Table 3 Convergence coefficient experiment

Weight configuration (α, β, γ, δ)	HD(mm)	ASSD(mm)
(0.25, 0.25, 0.25, 0.25)	12.7	3.86
(0.4, 0.2, 0.2, 0.2)	10.9	4.12
(0.1, 0.3, 0.3, 0.3)	8.3	2.74

To validate the rationality of the fusion coefficients (α, β, γ, δ) in Eq. (25), we designed three representative configurations under the constraint $\alpha+\beta+\gamma+\delta=1$: (0.25, 0.25, 0.25, 0.25), (0.4, 0.2, 0.2, 0.2), and (0.1, 0.3, 0.3, 0.3). The evaluation was conducted using two boundary and structural consistency metrics: Hausdorff Distance (HD) and Average Symmetric Surface Distance (ASSD), with results summarized in Table 3. Experimental findings reveal that equal-weight fusion fails to capture detailed edge structures, leading to considerable boundary deviation. Although increasing the global feature weight ($\alpha=0.4$) enhances long-range contextual modeling and slightly improves HD, it results in higher ASSD, indicating impaired boundary smoothness. In contrast, the configuration (0.1, 0.3, 0.3, 0.3) achieved the best performance across both metrics, demonstrating superior preservation of lesion contours and structural continuity. Therefore, this configuration was selected as the final fusion strategy to ensure optimal boundary alignment and spatial consistency.

3.3 CT image segmentation of liver cancer based on DconnNet

The DconnNet model is an advanced medical image segmentation network (Dai and Guan 2024), which is specifically designed to improve the accuracy of segmentation of visceral organs in CT images. The model effectively solves the segmentation difficulties such as blurred boundaries and complex morphology of visceral organs in CT images through unique directional connectivity modeling, and its advantages are mainly reflected in three aspects:

1) The continuity expression of organ topology is enhanced by using multidirectional convolution kernel;
2) Enhancing the localization accuracy of complex anatomical regions through dynamic feature calibration mechanism;
3) Constructing a lightweight coding and decoding architecture to realize high-resolution detail preservation.

Inheriting the above technical features and based on the DconnNet model, a NMD-FusionNet semantic segmentation model is designed in this study.

DconnNet mainly consists of two core modules: sub-path direction excitation Sub-path Direction Excitation (SDE) and Interactive Feature-space Decoder (IFD). The SDE module is responsible for extracting the direction subspace from the shared latent space through a series of operations to enhance the the representation of directional features to better capture the directional relationships among pixels. The IFD module, on the other hand, utilizes this directional information to fuse the directional features into the feature maps of each layer through interactive decoding streams, ensuring that the entire network can effectively utilize the direction-dependent features. In addition, DconnNet introduces a new loss function that specifically addresses the data imbalance problem commonly found in medical datasets, and improves the robustness and accuracy of segmentation by adjusting the weights based on the label size distribution. These innovations enable DconnNet to not only achieve higher segmentation accuracy and stability when dealing with complex liver tumor segmentation tasks, but

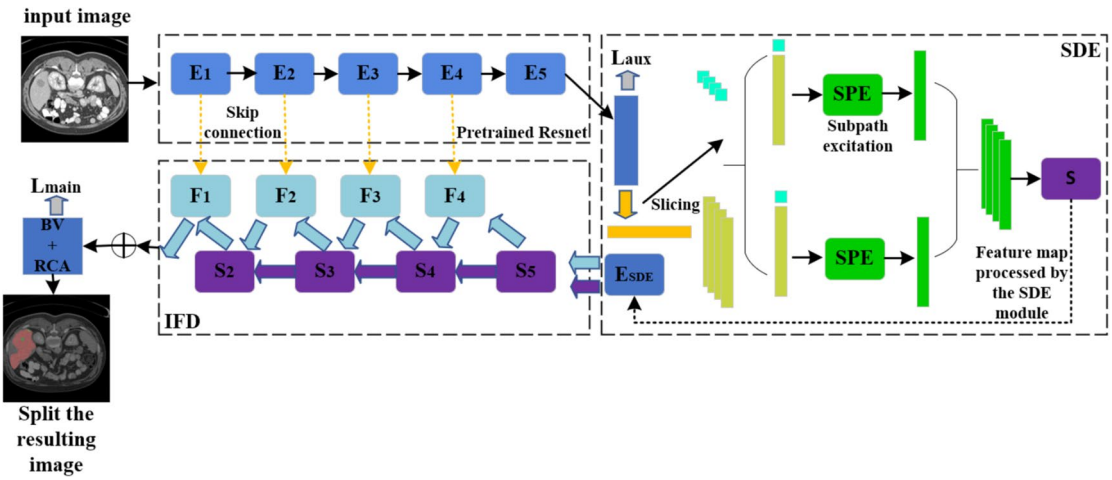


Fig. 4 DconnNet model framework diagram

also effectively deal with the complexity of the segmentation boundaries, significantly improving the ability to accurately depict liver tumor regions.

The overall structure of DconnNet is shown in Fig. 4. Direction connectivity-based segmentation includes direction a priori extraction, channel slicing, subpath excitation, interactive feature space decoder, and connectivity output, which are described in detail below.

By utilizing the channel orientation information in the connectivity mask, we obtain a unique orientation embedding. This is achieved by supervising the intermediate features and compressing the auxiliary connectivity output channels. At layer n, the output of the encoder is up-sampled to match the input size, resulting in a preliminary output that is trained to learn the connectivity mask. This preliminary output is then compressed by global average pooling and mapped to the same dimension as the latent feature map. To decouple from the category and orientation subspaces, we divide the latent feature and orientation prior into eight parts via channel slices. For each pair of feature embedding slices, we pass the feature slices through the spatial and channel attention modules in order to capture dependencies, resulting in updated feature slices. We multiply the orientation prior with the updated feature slices to highlight or suppress features based on specific orientation information, and then recode the output using a 1×1 convolution. We stack all subpath outputs and recode them into a new feature map.

The Interactive Feature Space Decoder (IFD) ensures that orientation-dependent information is efficiently fused into the feature maps in each layer. The IFD contains two dynamic top-down interactive streams: the spatial stream and the feature stream.Each layer of the IFD contains the main feature map and the orientation-enhanced map. In the first layer of the decoder, both maps are initialized to the output of the SDE. Each feature stream contains a feature block which includes two convolutional layers and an upsampling layer with jump connections. It takes the directionally enhanced map as input and outputs the up-sampled main feature map. The embedding of the spatial stream fuses the orientation information into the main feature map of the feature stream. In turn, the master feature map refines the embedding by combining the orientation information at higher resolution. Each set of eight channels of the master feature map represents a category of connectivity. The final segmentation map is obtained by a bilateral voting module and a region-guided channel aggregation module.

The total loss function (L_{total}) combines the primary loss (L_{main}) and the auxiliary loss (L_{aux}) from the directional connectivity module as follows:

$$L_{total} = L_{main} + 0.3 \times L_{aux} \tag{26}$$

where L_{main} denotes the loss of the main segmentation output, and L_{aux} denotes the loss of the auxiliary output of the directional connectivity module. Each loss component (L_{main} and L_{aux}) consists of two parts: the Class Density Loss (CDL, Class Density Loss) L_{cd} and the Connectivity Consistency Loss (CCL, Connectivity Consistency Loss) L_{cc} as follows:

$$L_{cT} = L_{cd} + L_{cc} \tag{27}$$

Category Density Loss (CDL) is used to solve the common problem of category imbalance in medical image datasets. It is based on the size distribution of the categories of interest in the dataset, which enhances the robustness and performance of the model in dealing with unbalanced datasets. Before training, we analyze all training images to generate the label size (total number of positive pixels) for each image. Based on this size distribution, we compute the probability density function of label size k for each category j, denoted as $PDF_j(k)$. During training, for each image, we compute the label size for each category j and derive its size density weight $P_j(k)$, as follows:

$$P_j(k) = \begin{cases} 1, & \text{if} k = 0 \\ -\log\left(PDF_j(k)\right), & \text{if} k \neq 0 \end{cases} \tag{28}$$

The category density loss is defined as an improved Dice loss with the following formula:

$$L_{cd} = \sum_j P_j(k)\left(1 - \frac{2\sum(S \times G) + \epsilon}{\sum S + \sum G + \epsilon}\right) \tag{29}$$

In Eq. S denotes the final segmentation prediction result, G denotes the true label, and ϵ is a stabilizing term that is usually set to 1.

The connectivity consistency loss (CCL) is used for connectivity modeling and consists of two components: edge consistency loss L_d and connectivity preservation loss L_c, defined as follows:

$$L_{cc} = L_d + L_c \tag{30}$$

The edge consistency loss (L_d) is used to ensure that the edges of the segmentation results are consistent with the edges of the real labels, avoiding broken or blurred edges; the connectivity preservation loss (L_c) is used to maintain the connectivity of the segmentation results, ensuring the integrity of the segmented region and avoiding isolated pixels or regions. By combining the category density loss and the connectivity consistency loss, the total loss function is able to optimize both segmentation accuracy and connectivity, thus improving the performance of the model in medical image segmentation tasks.

4 Experimentation

4.1 Dataset and evaluation indicators

The study dataset in this paper was provided by the Second People's Hospital of Huaihua City, and contained 2000 abdominal CT images from 400 patients, with 3–6 sequences acquired for each patient, including 300 positive cases of hepatocellular carcinoma (a total of 1,500 images, all of which contained clear focal areas) and 100 negative control cases (500 normal images). Three main imaging devices (GE Revolution CT 700 images, Siemens SOMATOM Drive 850 images, and Philips iCT 450 images) were used during data acquisition and covered three key time windows: the plain, arterial, and portal phases.

In the image pre-screening stage, this study used a screening algorithm (FSA) based on improved non-local mean filtering to accomplish the following screenings:

1) Full screening of negative images: among 500 negative images, all of them were rejected due to respiratory artifacts (320 images), intestinal gas interference (118 images) and equipment noise (62 images);
2) Positive image optimization: among 1500 positive images, low-quality images with severe motion artifacts (186 images), blurred lesion boundaries (94 images) and some significant volume effects (120 images) were eliminated, and 1200 high-quality lesion images were finally retained, with a screening retention rate of 80%.

In the image fusion stage, for the screened 1200 lesion images, enhanced images were generated by multidimensional complementary fusion:

1) Dynamic contrast across time windows: temporal and spatial alignment was performed for the swept phase (380 images), arterial phase (620 images), and portal phase (200 images) to generate 600 sets of fused images (2 dynamic sequences for each patient);
2) Multi-parameter optimization: Fusion of the window width and window position characteristics of the soft tissue window (480 sheets), liver window (420 sheets) and lung window (300 sheets) to generate 600 parametric fusion images;
3) Cross-device compensation: 600 heterologous compensation images were generated by aligning the feature space of three device images from GE (280 images), Siemens (520 images), and Philips (400 images).

The final 1200 high-precision lesion images were generated by adaptive weighted fusion, which was consistent with the number after screening. The average lesion contrast of the fused images was improved from 1.26 ± 0.15 before screening to 1.89 ± 0.21 ($p < 0.01$). The dataset was divided into a training set (840 images), a validation set (240 images) and a test set (120 images) in a 7:2:1 ratio. The deep learning image segmentation models in the subsequent comparison experiments used the lesion image datasets after screening and image fusion.

We chose a variety of assessment metrics, including intersection and merger ratio (IoU), Dice similarity coefficient (DSC), precision (Pre), recall (Re), and accuracy (Acc). We believe that the intersection and merger ratio (IoU) is one of the most important evaluation metrics.IoU can effectively represent the similarity between the actual tumor region and the predicted tumor region. Similar to IoU, Dice coefficient (DSC) is also used to express the similarity.Pre describes the situation where non-tumor regions are incorrectly segmented into tumor regions, while Re represents the proportion of correctly predicted tumor regions to manually annotated tumor regions.Acc represents the proportion of correctly predicted samples among all samples.

4.2 Experimental details

The training of the model was performed on a server equipped with NVIDIA RTX 4090 GPU, implemented using the Pytorch framework. We used the Adam optimizer with an initial learning rate of 0.00004. training the semantic segmentation model ran for 100 cycles. To ensure the efficiency and robustness of our proposed model in dealing with the task of segmenting CT images of liver cancer, we used three strategies including data enhancement, model structuring and training techniques.

Data enhancement is a key technique to improve the generalization ability of the model. For the liver cancer CT image dataset, we applied various data enhancement techniques including rotation, flipping, scaling, panning, and color dithering. These techniques increase the diversity of training samples and reduce model overfitting to specific sample patterns. For example, we generate new training samples by randomly rotating images from 0 to 360 degrees, performing horizontal and vertical flips, and applying random scaling and cropping.

Regarding model structural adjustment, we implemented a dynamic adjustment mechanism during the model training process to improve the model adaptability and training efficiency. For example, depending on the training stage, we dynamically adjust the depth and width of the network to ensure that the model can quickly learn basic features in the early stage and deeply mine complex features in the later stage. In addition, we also dynamically adjust the attention weights according to the changes in data distribution so that the model can pay more attention to the key

information in the data. This dynamic adjustment strategy not only improves the training efficiency of the model, but also enhances the generalization ability of the model on complex or changing datasets.

In terms of training techniques, we perform color dithering by adjusting brightness, contrast and saturation to further enrich the training data. In addition, during the training process, we apply several practical training techniques to enhance the model performance. For example, we use the early-stop method to stop training when the performance on the validation set is no longer improving, thus preventing model overfitting.

4.3 Comparative experimental results and analysis

In order to verify the effectiveness of the models in this paper, we have selected six models that perform well in the general area of semantic segmentation as baseline models for comparison experiments, which are described as follows:

1) U-Net (Niu et al. 2024): A classical image semantic segmentation model that combines a symmetric encoder-decoder architecture with cross-layer hopping connections, after capturing contextual features through downsampling, local details and global semantic information are fused by upsampling and hopping connections.
2) HRNet (Wang et al. 2025c): A high-resolution maintained vision model, which maintains high-resolution representations throughout the process through parallel multiresolution subnetworks and continuous multiscale feature interactions, effectively preserving detail information and significantly improving semantic segmentation accuracy.
3) Swin-unet (Cai and Zhang 2025): A pure Transformer model for image segmentation, combining U-shaped architecture with shift-window self-attention mechanism to improve lesion localization accuracy through global context modeling and cross-resolution feature fusion.
4) AttU-Net (Gu et al. 2025): An improved image segmentation model that introduces an attention gating mechanism on the basis of U-Net to enhance the localization and segmentation accuracy of complex lesions by dynamically focusing on key regions in multi-scale features and enhancing the feature response of the target region.
5) TransUNet (Wang et al. 2025a): A hybrid architectural model for image segmentation that combines the global context modeling capability of Transformer with the local feature extraction and jump connections of U-Net, enhances cross-scale semantic fusion through visual-linguistic pre-training in the encoder, and significantly improves the segmentation accuracy of complex images.

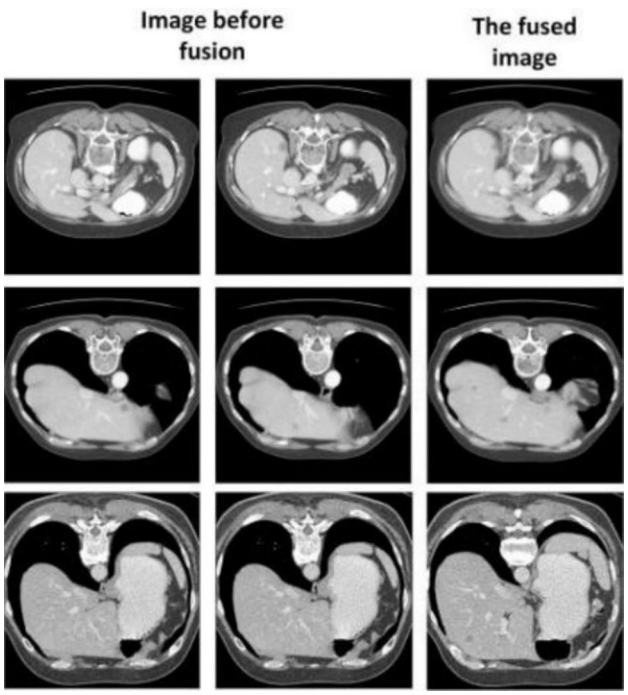


Fig. 5 Example of image fusion

6) ResUNet (Hu et al. 2025): An improved image segmentation model that embeds residual connection module in U-Net architecture, which improves deep network training stability and lesion segmentation accuracy by alleviating the gradient vanishing problem with enhanced feature reuse capability.

Figures 5 and 6 below show specific examples of MSFF image fusion and the comparison of segmentation results of DconnNet model before and after image fusion, respectively. The two images on the left side of Fig. 7 represent the original CT images acquired at different time points with different imaging parameters and different imaging devices, while the image on the right side is the result after fusion, which visualizes how the MSFF method effectively integrates the information from multiple sources to generate fused images with better quality. The fused images show significant advantages in noise suppression, detail preservation and contrast enhancement. Specifically, the MSFF method is able to effectively reduce random noise, improve signal-to-noise ratio, and make the lesion boundary clearer through multi-scale feature fusion filtering. In addition, the method enhances the local high-frequency information, avoids the loss of details, and optimizes the texture and structure representation of the lesion region. Meanwhile, the fused image enhances the gray contrast between the lesion and the background tissue, making it easier to identify the lesion in the low-contrast region, thus providing reliable image support for subsequent image segmentation.

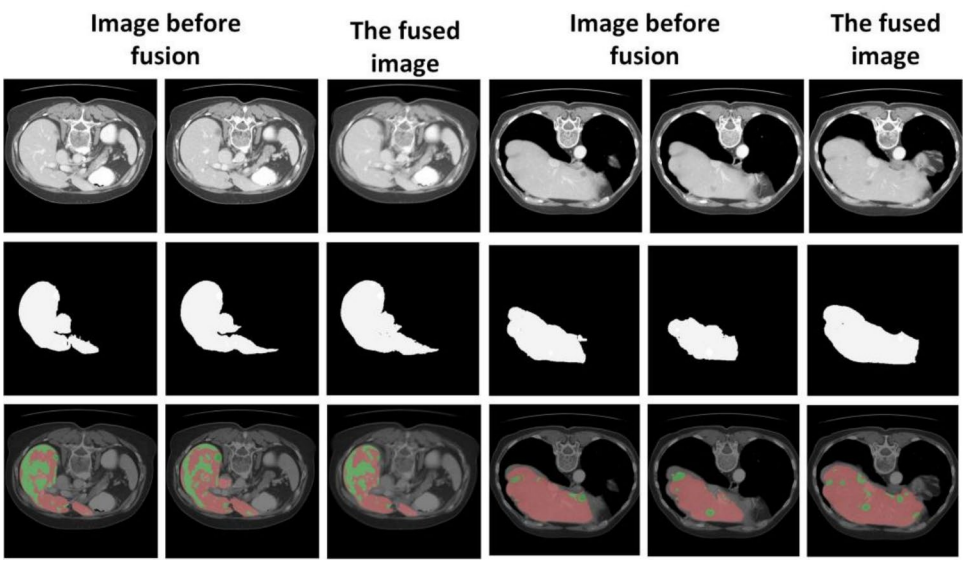
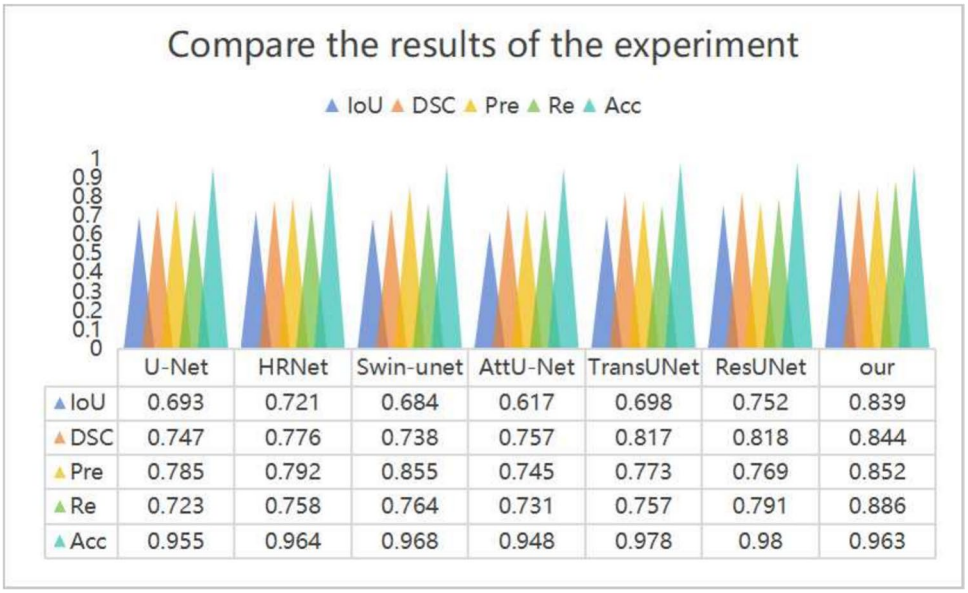


Fig. 6 Comparison of segmentation results before and after image fusion



	U-Net	HRNet	Swin-unet	AttU-Net	TransUNet	ResUNet	our
IoU	0.693	0.721	0.684	0.617	0.698	0.752	0.839
DSC	0.747	0.776	0.738	0.757	0.817	0.818	0.844
Pre	0.785	0.792	0.855	0.745	0.773	0.769	0.852
Re	0.723	0.758	0.764	0.731	0.757	0.791	0.886
Acc	0.955	0.964	0.968	0.948	0.978	0.98	0.963

Fig. 7 Comparison of experimental results

Figure 6 further verifies the improvement of image fusion on tumor segmentation performance. The image before fusion is susceptible to noise interference due to the limitation of a single imaging condition, resulting in blurred lesion boundaries and affecting the accuracy of the segmentation results in terms of lesion contour portrayal and detail extraction. After fusion, the image quality was significantly improved, with enhanced contrast and clearer edges in the lesion region, and the model's ability to capture lesion morphology was enhanced. In addition, the fused image shows higher sensitivity in small tumor detection, which can effectively reduce the risk of missed detection, improve the overall segmentation accuracy, and provide a more reliable imaging basis for accurate tumor diagnosis.

Table 4 Comparison of experimental results

Model	IoU	DSC	Pre	Re	Acc
U-Net	0.693	0.747	0.785	0.723	0.955
HRNet	0.721	0.776	0.792	0.758	0.964
Swin-unet	0.684	0.738	0.855	0.764	0.968
AttU-Net	0.617	0.757	0.745	0.731	0.948
TransUNet	0.698	0.817	0.773	0.757	0.978
ResUNet	0.752	0.818	0.769	0.791	0.980
our	0.839	0.844	0.852	0.886	0.963

In this section, experiments are carried out on the CT image dataset of hepatocellular carcinoma using filtered and image fused images, and the results are compared with the existing baseline model results, which are shown in Table 4 and Fig. 7 below.

The proposed model significantly outperforms traditional methods in several evaluation indexes, mainly due to the synergistic optimization of the non-local threshold screening algorithm, the multi-scale feature fusion framework and the DconnNet architecture. First, the nonlocal threshold screening algorithm preprocesses the original image to effectively suppress noise interference and local gray scale unevenness, and dynamically adjusts the threshold value to enhance the contrast between the target region and the background, thus improving the robustness of segmentation. The method performs well in Precision (85.2%) and Recall (88.6%), indicating that it is able to reduce mis-segmentation while preserving the target anatomical structure more completely.

Second, the multiscale feature fusion mechanism improves the model's ability to retain boundary details and recognize complex structures by aggregating information from different sensory fields in parallel, resulting in improved IoU (83.9%) and DSC (84.4%). In addition, DconnNet's directional connectivity modeling further optimized the structural consistency of the target region, reduced the fragmentation segmentation problem, and maintained a high level of Accuracy (96.3%), which effectively reduced the risk of missed detection of small lesions.

Compared with traditional models such as U-Net, this method realizes the whole process enhancement through the three-stage optimization of "preprocessing-feature expression-structural constraints". Among them, non-local threshold screening provides high-quality input, multi-scale fusion compensates for the limitation of single-network receptive fields, and the directional connection mechanism of DconnNet guarantees the anatomical rationality of segmentation results. This innovative design enables the model to outperform HRNet (11.8% IoU enhancement) and TransUNet (2.7% DSC enhancement) on the liver cancer CT dataset, demonstrating higher segmentation accuracy and robustness, and providing an effective solution for medical image analysis.

Figure 8 demonstrates a significant advantage at the precision-recall (PR) point, achieving up to 0.852 precision and a commendable 0.886 recall. Compared to other models such as Swin-Unet (precision of 0.855 but recall of only 0.764) and HRNet (precision of 0.792 and recall of 0.758), our model ranks ahead. It effectively balances precision and recall in capturing positive samples, demonstrating excellent overall performance.

Figure 9 below demonstrates multiple actual segmentation results in the CT image dataset of hepatocellular carcinoma using filtered and image fused images, in order to visually compare the segmentation performance of different methods. From the results, it can be seen that the method proposed in this paper performs well in boundary inscription and detail preservation of the lesion region, and is able to accurately capture the morphological features of the lesion, especially in the case of blurred tumor edges or low contrast, and still maintains high segmentation accuracy. In addition, compared with other comparison models, the method in this paper has an obvious advantage in the identification of small tumors, which can effectively detect lesions with small size and irregular morphology, reducing the risk of missed detection. This performance enhancement is mainly due to the synergistic effect of non-local threshold lesion screening and multi-scale feature fusion strategy, which provides the model with stronger robustness and generalization ability in complex backgrounds. Comprehensively, the experimental

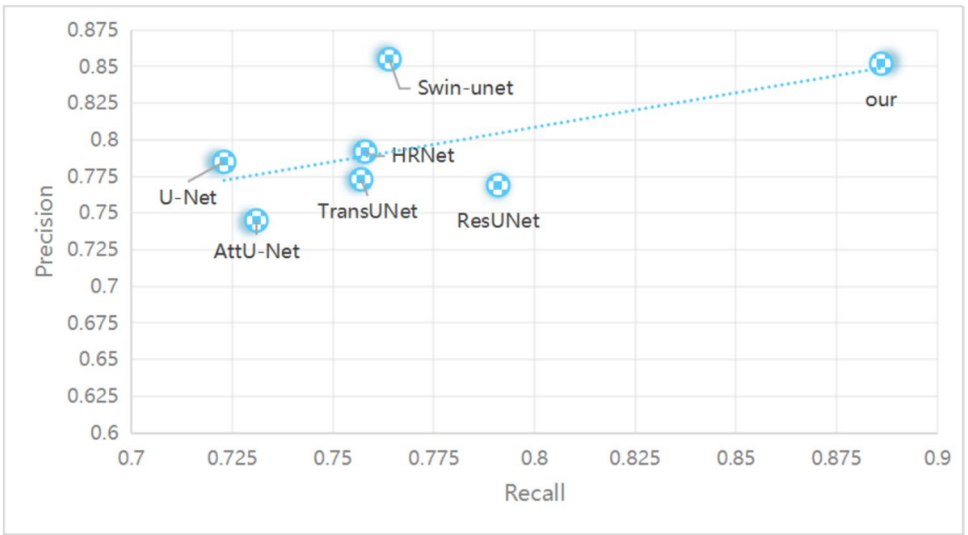


Fig. 8 Precision-recall points for different models

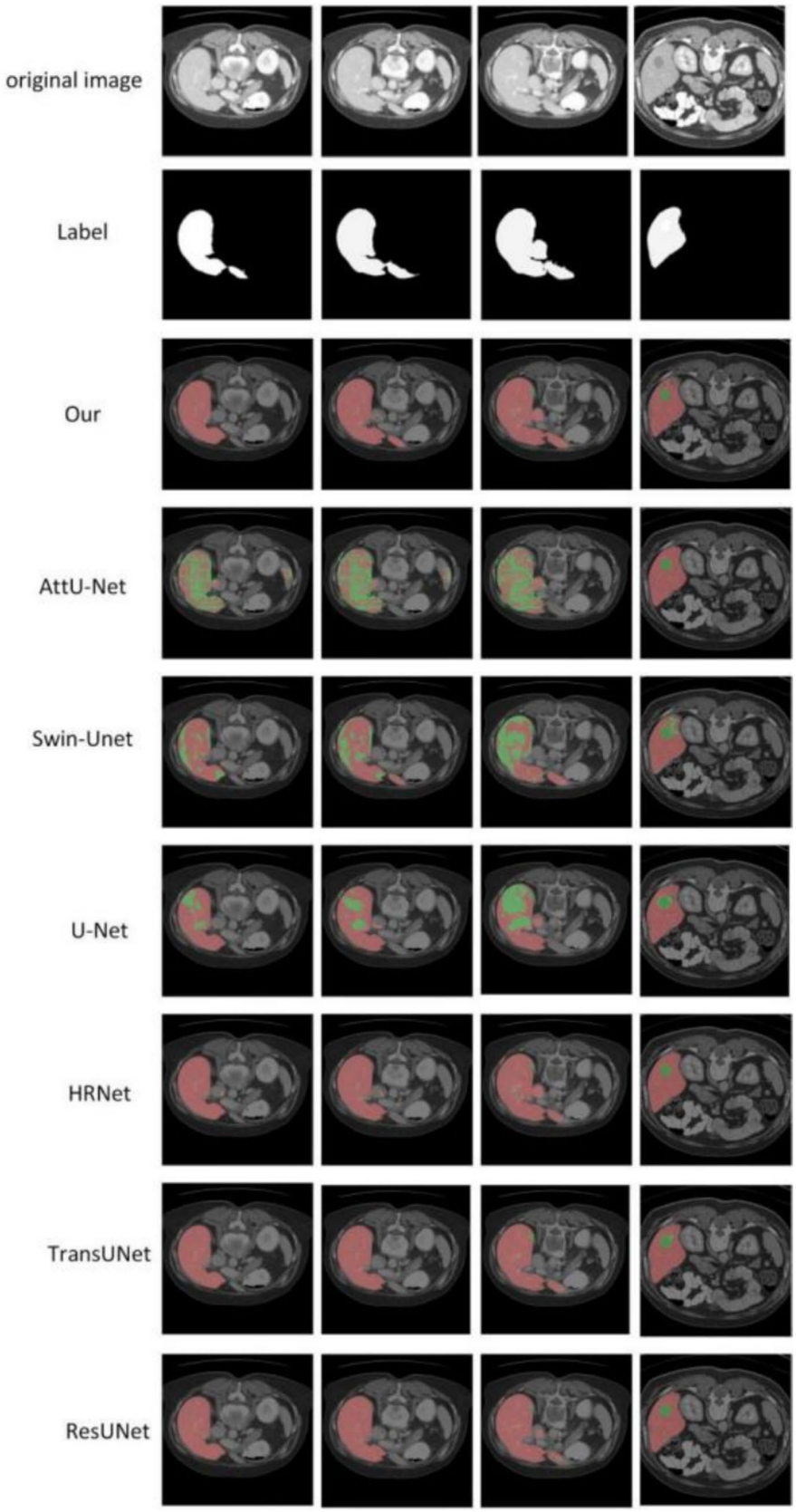


Fig. 9 Comparison of test results using different methods

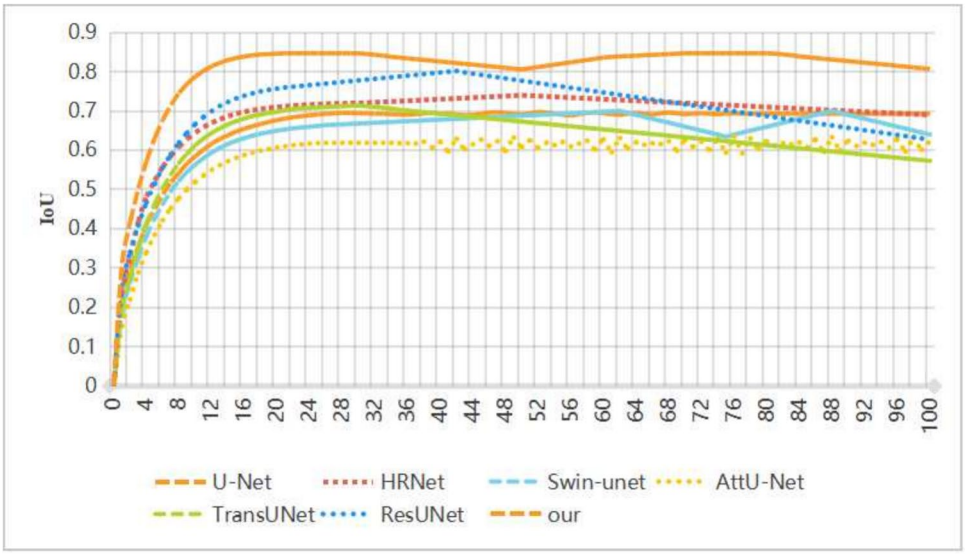


Fig. 10 The change in IoU metric during the training of 100 epochs

results further validate the effectiveness of this paper's method in the task of CT image segmentation of liver cancer, especially in dealing with low-contrast, small-target foci, and complex morphological changes, with better segmentation results compared with existing methods.

Figure 10 below shows the evolution of the IoU index for NMD-FusionNet and its comparison models over the course of 100 training cycles. As can be seen from the figure, all the equivalent models have much lower IoU index values than NMD-FusionNet in most of the cases. As the training process continues, the values of NMD-FusionNet stabilize and show an overall increasing trend without overfitting, while this is not the case for some of the comparative models, which show a decreasing trend after a period of stabilization, i.e., there is an overfitting phenomenon. i.e., there is an overfitting problem.

We further conducted a comprehensive comparison of the proposed method with three representative mainstream models-ConvNext (Woo et al. 2023), Mamba(Liu et al. 2024), and SAM (Sun et al. 2024). As can be seen from the Fig. 11, the performance of our model (OUR) is more stable across the 20 sets of samples, with its DSC metrics mostly maintained above 0.80 with a small range of fluctuations, which indicates that the model has good

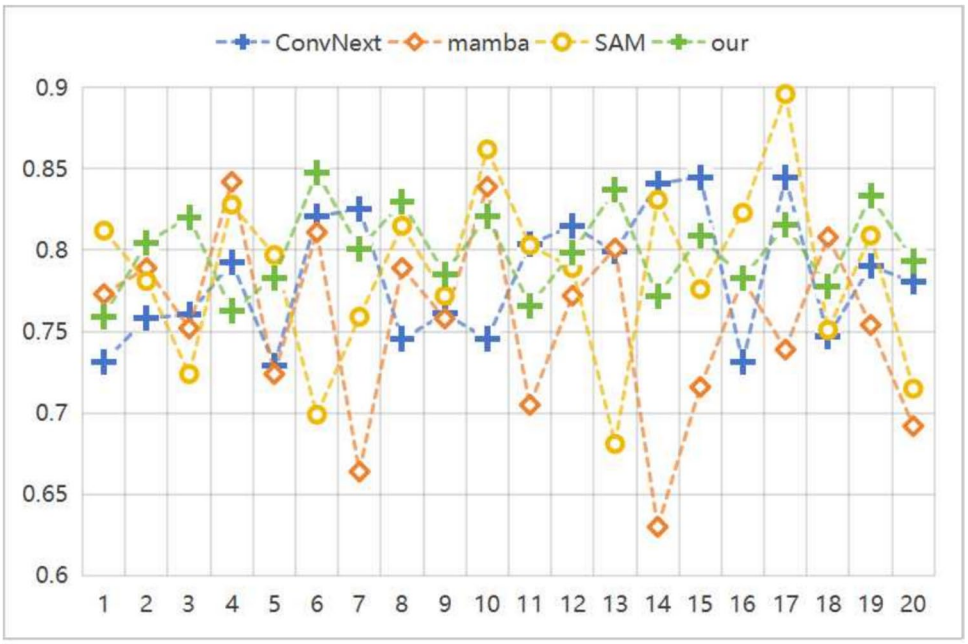


Fig. 11 Comparison of the performance of our proposed model with three newer models on random samples

robustness. In contrast, the performance of the Mamba model fluctuates more, with DSC values below 0.75 at several sample points, and the lowest is even close to 0.65, showing that its performance is unstable when dealing with some complex images.The performance of ConvNext is low overall, with most values clustered around 0.74, while the SAM model, although it achieves higher DSCs in several samples (e.g., group 16), the overall distribution is still slightly inferior. Taken together, our method not only has a high average performance, but also maintains a small standard deviation across samples, showing stronger generalization ability and practical potential, especially for the task of liver tumor medical image segmentation, which requires stable performance output.

4.4 Ablation experiments

In order to deeply verify the effectiveness of the non-local thresholding lesion-based image screening algorithm and the image multi-scale feature fusion algorithm, this study systematically analyzes the model performance through ablation experiments. The experimental results are shown in Table 5 and Fig. 12. our1 is the DconnNet base model, our2 is DconnNet + FSA, our3 is DconnNet + MSFF, and our is our proposed method NMD-FusionNet. the DSC of the original DconnNet semantic segmentation model (our1) reaches 0.796, which initially validates the effectiveness of the infrastructure. After introducing the nonlocal threshold-based pathology image screening algorithm (our2) on this basis, the DSC is significantly improved to 0.862. Unlike the traditional NLM algorithm which only relies on the original pixel/grayscale space for similarity computation, the FSA acts on the deep feature space and introduces a dynamic feature channel attention mechanism. The algorithm not only effectively suppresses the noise interference that traditional NLM is susceptible to in complex pathological backgrounds, but also significantly improves the model's ability to recognize subtle lesion features and robustness to background complexity by selectively enhancing the feature channels that are most relevant to key pathological structures. This result highlights the key role of FSA in accurate extraction and enhancement of pathology features.

After fusing the image multiscale feature fusion algorithm (our3) on top of our1, the DSC is further improved to 0.864, and the Pre metric is optimized simultaneously to 0.882. Multiscale feature fusion improves the segmentation accuracy of the model in the edge blurring region by integrating the local details with the global semantic information, which increases the accuracy of the model in the edge blurring region by 2.1%, and shows stronger feature capturing ability especially in the low-contrast pathology slides. The final model (OUR) integrating the complete optimization scheme has the best overall performance, with Iou reaching 0.839, which is a 9.6% improvement over the original model. The experimental results show that the synergy of non-local threshold screening and multi-scale fusion

Table 5 Ablation of experimental results

Model	IoU	DSC	Pre
our1	0.743	0.796	0.836
our2	0.765	0.862	0.843
our3	0.758	0.864	0.882
our	0.839	0.844	0.852

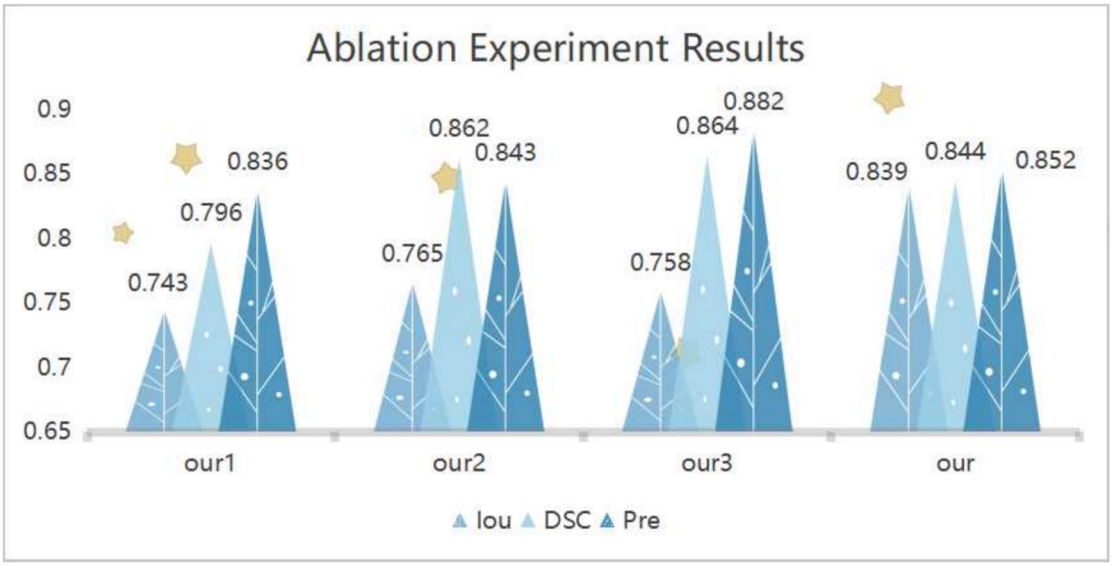


Fig. 12 Ablation experiment results

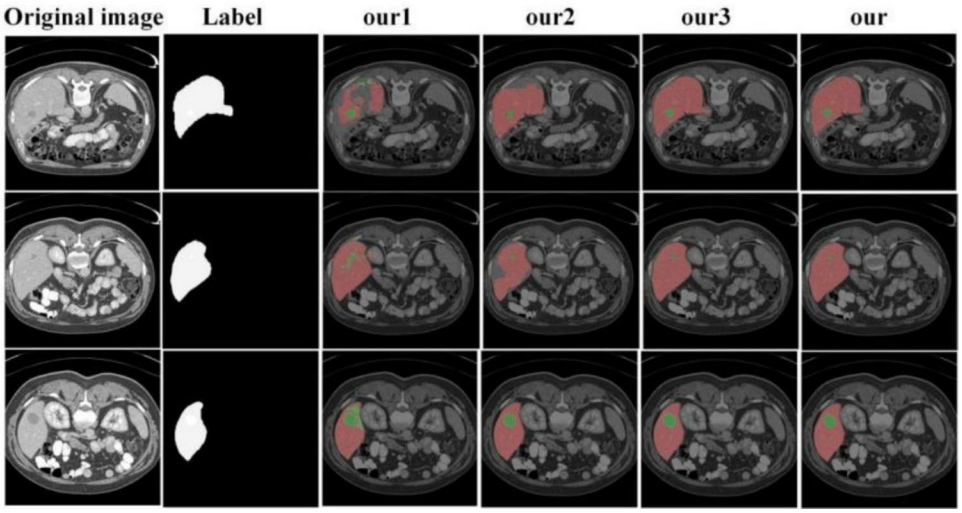


Fig. 13 Comparison of segmentation results of different models in ablation experiments

effectively solves the challenges of noise sensitivity and feature scale differences in pathology images, and verifies the rationality of the algorithm design.

Figure 13 demonstrates the specific performance of different models of ablation experiments in the task of CT image segmentation of hepatocellular carcinoma to visually verify the effectiveness of the proposed method. The image consists of multiple rows and columns, which sequentially compare the ability of different methods in lesion region identification. The first column shows the original CT scan image as the input data source. The second column gives the manually labeled gold standard (Ground Truth), i.e., the lesion regions manually outlined by the expert, for evaluating the segmentation accuracy of the model. The third and subsequent columns show the segmentation results of different models in turn.

From the experimental results, the segmentation results of our1 (DconnNet base model) have some errors, some lesion regions are not accurately recognized and there is a certain degree of mis-segmentation. After the introduction of the non-local threshold lesion screening algorithm (our2), the segmentation effect of the model in the complex background is improved, the contour of the lesion region is clearer, and the phenomenon of mis-segmentation is obviously reduced. After combining the image multi-scale feature fusion (our3) on the basis of our1, the segmentation accuracy of the model in the boundary region is further optimized, showing better lesion region alignment ability. Ultimately, the fully optimized our model demonstrates the best segmentation performance, with segmentation results that highly match the gold standard and the least number of mis-segmented regions, proving the effectiveness of the synergy between

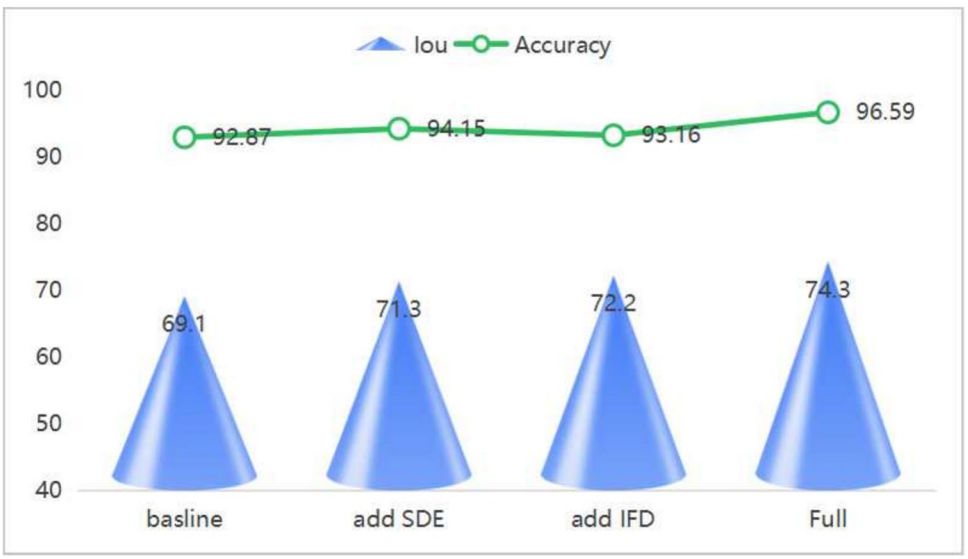


Fig. 14 Comparison of the performance of each module of the segmentation model

non-local threshold screening and multi-scale feature fusion in enhancing the robustness of the model.

In order to verify the role of two key modules, subpath direction excitation (SDE) and interactive feature space decoder (IFD), in the DconnNet segmentation model, the following ablation experiments are designed, and the results are shown in Fig. 14. The Basline model architecture is a resnet, and SDE and IFD are added to Basline, respectively. The experiments show that the subpath direction excitation (SDE) and the Interactive Feature Spatial Decoder (IFD) module have significant synergistic effects on segmentation performance improvement: when only the SDE module is introduced, it enhances pathological structural features through orientation-sensitive convolution, resulting in an increase of IoU by 0.022%; deploying the IFD module alone optimizes the boundary recognition by virtue of the cross-scale feature interaction mechanism, driving an increase of Accuracy by 0.29%; the dual modules in combination produce a super-additive gain, the IoU of the complete model (SDE + IFD) reaches 74.3%, and its gain amplitude exceeds the sum of the contributions of the independent modules, which proves that the directional feature encoding capability of SDE and the spatial context fusion mechanism of IFD effectively complement each other in solving the structural fragmentation and boundary ambiguity problems of pathology images.

4.5 Comparative methodological analysis with representative models

To further clarify the source of performance improvements, we compare our NMD-FusionNet with representative models such as UNet and HRNet from a methodological perspective.

Architectural Design: UNet uses a symmetric encoder-decoder with skip connections, but its fixed-scale structure limits its capacity to handle multi-phase CT features. HRNet maintains high-resolution features but lacks explicit mechanisms to suppress background noise. In contrast, our dual-path DconnNet architecture incorporates directional excitation and spatial awareness modules, enhancing boundary precision and feature reuse.

Feature Extraction: UNet and HRNet rely on standard convolutional hierarchies and do not address motion artifacts or phase fusion. Our method introduces an improved non-local filtering module (FSA) and a multi-scale fusion strategy, enabling robust feature integration across devices and acquisition phases.

Training Strategy: Unlike typical setups that use cross-entropy or Dice loss, we employ a boundary-constrained loss function and include adaptive pre-screening and denoising, improving convergence and segmentation accuracy.

In summary, NMD-FusionNet achieves superior results not only due to strong performance metrics but also because of tailored innovations that address the challenges of liver tumor CT segmentation in both architecture and training.

5 Conclusion

To address the efficiency bottlenecks and accuracy challenges in liver cancer CT image segmentation, this study proposes a novel three-phase processing framework comprising pre-screening, fusion denoising, and deep segmentation. The method integrates non-local mean filtering for data purification, a multi-scale feature fusion denoising algorithm to preserve fine structures, and an improved DconnNet model incorporating multi-level skip connections and dynamic convolution modules. Experimental results demonstrate that our method achieves an 83.9% IoU in complex edge segmentation tasks and outperforms benchmark models like ResUNet by 8.3% in accuracy, significantly reducing the manual review burden and offering robust support for clinical decision-making systems.

However, the current approach has several limitations. The dataset is primarily composed of enhanced CT scans from a single medical center, limiting the generalizability to heterogeneous modalities such as ultrasound or multiparametric MRI. Additionally, the adaptive thresholding in the pre-screening stage requires iterative computation, which hampers real-time performance on low-end hardware. Furthermore, the method is currently designed for static images, with limited ability to capture the spatio-temporal dynamics inherent in 4D CT data.

Future work will focus on three directions: constructing a multicenter, cross-modal joint training framework to enhance the model's generalization across diverse imaging modalities; developing a lightweight, edge-computing-compatible version to support real-time segmentation in resource-constrained environments; and exploring the integration of segmentation results with clinical biochemical indicators to support intelligent, multi-dimensional diagnosis. These efforts aim to shift hepatocellular carcinoma diagnosis from traditional morphological analysis to data-driven, intelligent decision-making, and the proposed technical route offers valuable insights for advancing precision oncology in the era of smart medicine.

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Data availability All data analyzed during the current study are included in the submission. Data used to support the findings of this study are currently under embargo while the research findings are commercialized. Requests for data, 12 months after publication of this article, will be considered by the corresponding author.

Declarations

Use of AI tools declaration The authors declare we have not used Artificial Intelligence (AI) tools in the creation of this article.

Conflict of interest The authors declare no conflicts of interest.

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